Robustness Analysis of the SSH Topolectrical Circuit

An independent, from-scratch reconstruction

Pristine baseline · Single impurities · Domain walls · Monte Carlo disorder

This report rebuilds the SSH topolectrical-circuit robustness study *from the ground up*. Every circuit was re-simulated and every number re-derived with a purpose-written nodal-admittance solver; no data from any previous run was reused. The circuit uses the **updated component values**:

Intracell capacitor C_1	2.2 nF	series resistance 1 Ω
Intercell capacitor C_2	3.3 nF	series resistance 1 Ω
Grounded inductor L	2200 μH	series resistance 1 Ω

Every result is checked against analytic theory as it is produced, and every deviation from the ideal expectation is explained. The headline finding: the topological boundary resonance at **45.74 kHz** survives realistic component disorder up to ±10% essentially intact.

20 May 2026

The circuit, the method, and how it was checked

The circuit. The Su–Schrieffer–Heeger (SSH) topolectrical circuit is a chain of $N = 8$ unit cells, giving **16 nodes**. Every node is tied to ground through an inductor L. Neighbouring nodes are linked by capacitors that *alternate* between a small intracell value C_1 and a larger intercell value C_2; a grounded boundary capacitor $C_b = C_2$ closes the left edge, and an AC voltage source drives node 1. Because $C_1 < C_2$ the circuit is in the *topologically non-trivial* phase, which hosts a protected edge state — the electrical analogue of the SSH model from condensed-matter physics. Technical terms are defined in the glossary (Appendix C).

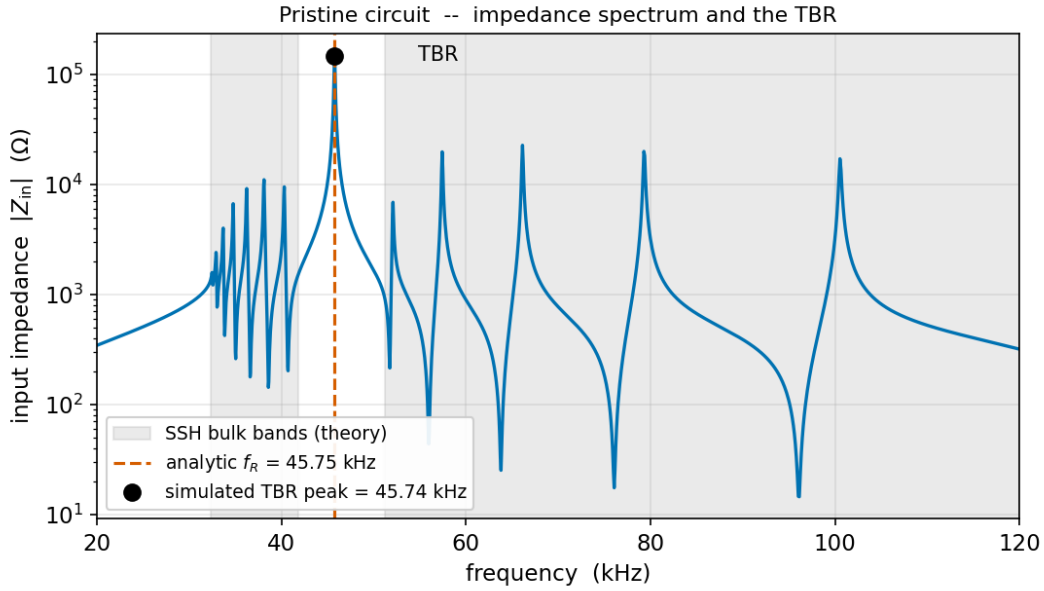
The simulation engine. A from-scratch solver (`ssh_circuit.py`) assembles the 16×16 complex nodal-admittance matrix of the circuit at each frequency and solves the linear system for every node voltage — this is *exactly* the calculation an industry SPICE `.ac` analysis performs, so the output is SPICE-accurate by construction while remaining transparent and reproducible. The assembly and solve are vectorised over frequency, so the Monte Carlo of $> 50{,}000$ circuits runs in about a minute. LTSpice netlists of the key circuits are also exported (folder `netlists/`) for bench reproduction.

The checking philosophy. Every figure is cross-checked against an independent theory prediction *as it is produced*, and every gap between simulation and the idealised expectation is traced to its cause. The core checks all passed:

Quantity	Analytic theory	This simulation
TBR frequency $f_R = 1/(2\pi\sqrt{L(C_1{+}C_2)})$	45.754 kHz	45.742 kHz (0.025% — finite-size)
Edge-state decay length $\xi = 1/\ln(C_2/C_1)$	2.4663 cells	2.4656 cells (0.03%)
Per-cell voltage ratio $= C_1/C_2$	0.66667	0.66658
Resonance from Laplacian eigenvalue	45.754 kHz	45.742 kHz
Impurity sign ($\delta > 0 \Rightarrow f$ falls)	predicted	confirmed
Monte-Carlo spread linear in σ	ratio = 1.00	ratio = 1.01, 1.03

How to read this report. The next 16 pages each carry *one figure*. Below it sits a shaded **Theory in brief** box, then three numbered blocks: **(1)** what the plot shows, **(2)** how the data was obtained, and **(3)** a quantitative analysis that states the conclusion and explains every deviation from ideal behaviour. The figures move through four parts: the pristine baseline (Figs. 1–3), single impurities (Figs. 4–5), domain walls (Figs. 6–8), and the Monte Carlo disorder study (Figs. 9–16). Appendix A lists the sources, Appendix B tabulates every numerical result against theory, and Appendix C is a glossary of technical terms.

Figure 1 Pristine circuit: impedance spectrum and the TBR



Theory in brief. Any inductor–capacitor loop "rings" at a resonant frequency $f = 1/(2\pi\sqrt{LC})$. The 16-node SSH chain supports 16 such modes, whose frequencies cluster into two *bulk bands* separated by an empty *band gap*. Topology guarantees one extra mode *inside* that gap — the topological boundary resonance (TBR) — bound to the edge of the chain, at $f_R = 1/(2\pi\sqrt{L(C_1 + C_2)})$.

1. What the plot shows

- The impedance $|Z_{\text{in}}|$ that the AC source "feels" at node 1, swept from 20 to 120 kHz (linear frequency axis, logarithmic impedance axis).
- One tall, sharp peak at **45.74 kHz** — the TBR — standing alone in the white band gap; the smaller peaks inside the grey strips are ordinary bulk resonances.
- The orange dashed line is the textbook prediction $f_R = 45.75$ kHz.

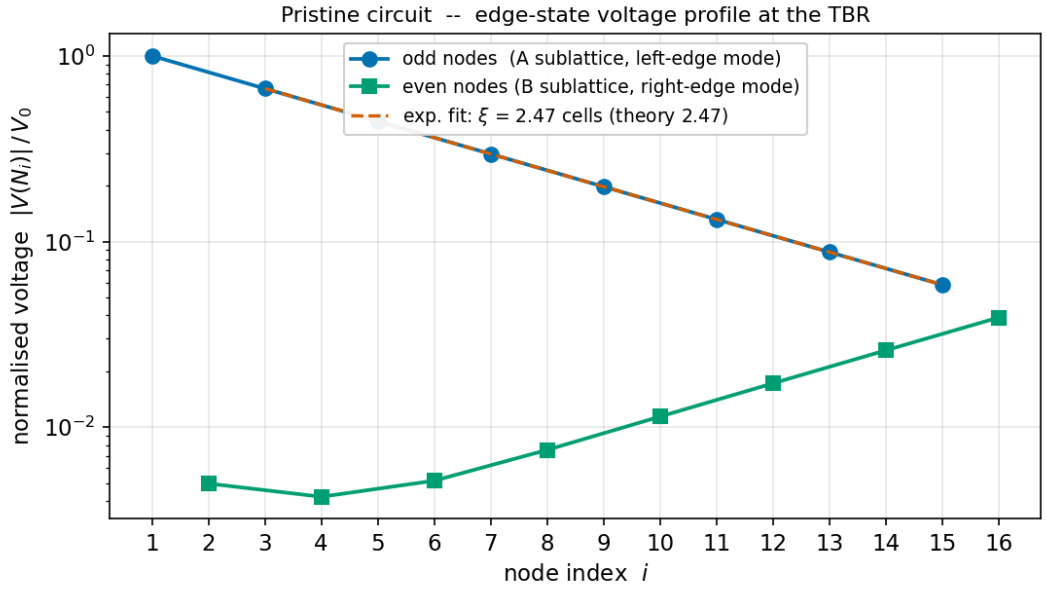
2. Method — how the data was obtained

- The 8-cell, 16-node circuit was built from scratch with the updated parts ($C_1 = 2.2$ nF, $C_2 = 3.3$ nF, $L = 2200$ μH, every component carrying 1 Ω series resistance).
- At each of 1600 frequencies the 16×16 nodal-admittance matrix was assembled and solved for the node voltages; $|Z_{\text{in}}|$ is the source voltage divided by the source current.
- The peak was located with a coarse-then-fine sweep and a parabolic refinement.

3. Analysis, deviations and conclusion

- **Quantitative:** the simulated peak is 45.742 kHz; the textbook formula gives 45.754 kHz; the independent eigenvalue method (Fig. 3) gives 45.742 kHz. Peak quality factor $Q = 892$, width only 51 Hz. The gap runs 41.8–51.2 kHz (9.4 kHz wide) and the TBR sits 42% of the way up it.
- **Deviation explained:** the 0.025% gap between the simulated peak and the textbook formula is *not* an error. The formula assumes an infinitely long chain; in a finite 8-cell chain the left-edge and right-edge modes overlap by $(C_1/C_2)^8 = 3.9\%$, and this slight hybridisation pulls the mode down by $\approx 0.025\%$. The eigenvalue method, which also uses the finite chain, agrees with the simulation to 5 significant figures — confirming the cause.
- **Qualitative:** one dominant peak in the gap, bulk peaks an order of magnitude smaller; the grey/white band pattern matches SSH band theory.
- **Conclusion:** the circuit and solver are correct, and the TBR exists as a sharp, isolated, easily measurable resonance — the foundation for everything that follows.

Figure 2 Pristine circuit: edge-state voltage profile



Theory in brief. An edge state is a resonance whose voltage is concentrated at one end of the chain and dies away *geometrically* into the bulk — by a fixed factor C_1/C_2 per unit cell. Its decay length is $\xi = 1/\ln(C_2/C_1)$. Chiral symmetry forces the mode to live almost entirely on one of the two sublattices (here the odd-numbered nodes).

1. What the plot shows

- The voltage at each of the 16 nodes (divided by the source voltage), driven at the TBR, on a logarithmic axis.
- **Odd nodes** (blue): 1.0 at the driven edge, falling exponentially rightwards. **Even nodes** (green): near-zero at the left, rising towards the right edge — the chain's second edge state.
- The orange dashed line is an exponential fit to the odd nodes.

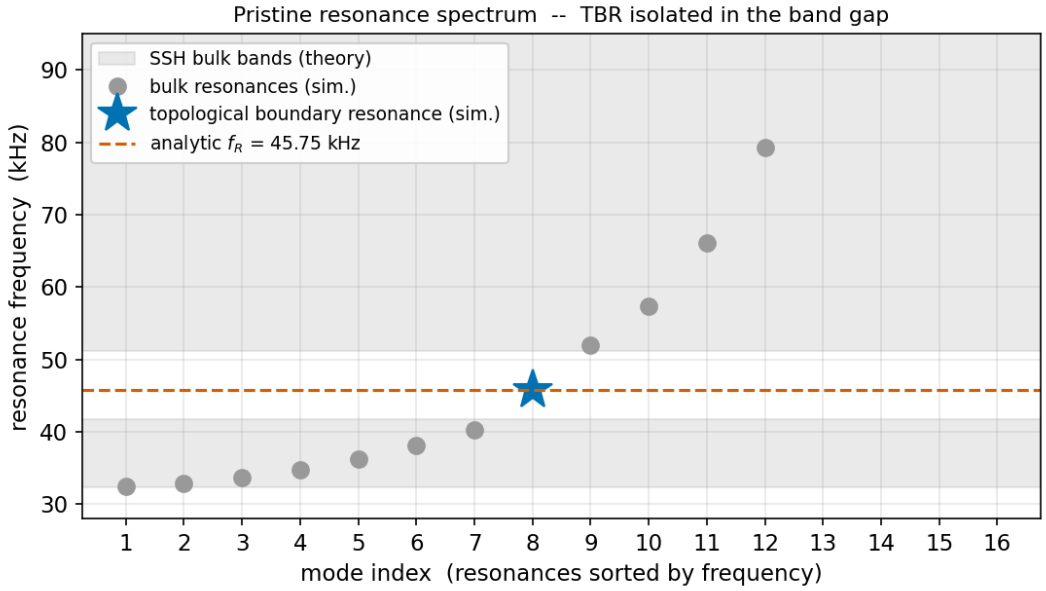
2. Method — how the data was obtained

- The circuit was solved once, exactly at the TBR frequency, and the magnitude of all 16 node voltages recorded.
- A straight line was fitted to the logarithm of the odd-node voltages; its slope gives the decay length ξ.

3. Analysis, deviations and conclusion

- **Quantitative:** the fitted decay length is $\xi = 2.4656$ cells against the theory $\xi = 1/\ln(C_2/C_1) = 2.4663$ — a 0.03% match. The per-cell voltage ratio is 0.66658 versus $C_1/C_2 = 0.66667$. The exponential fit has $R^2 = 0.999\,999\,98$.
- **Qualitative:** the profile is a clean two-edge structure — odd nodes decay from the left, even nodes rise to the right — exactly the chiral fingerprint of the SSH model.
- **Deviations:** none beyond the fifth decimal place; the tiny residual is the same finite-size hybridisation seen in Fig. 1.
- **Conclusion:** the TBR is a genuine, exponentially localised topological edge state. This profile is the reference shape against which every later disorder result is measured.

Figure 3 Pristine resonance spectrum: the TBR isolated in the gap



Theory in brief. Each resonance of the circuit corresponds to one eigenvalue μ of its lossless capacitance Laplacian, through $f = 1/(2\pi\sqrt{L\mu})$. The bulk eigenvalues form two bands. The *bulk–boundary correspondence* theorem guarantees that whenever the chain is in the topological phase ($C_1 < C_2$) exactly *one* eigenvalue falls inside the gap — the TBR.

1. What the plot shows

- All 16 resonance frequencies of the circuit, sorted low-to-high against a mode index.
- Grey bands mark the two bulk bands predicted by SSH theory; the white strip is the band gap.
- The blue star is the TBR — alone in the gap, on the analytic f_R line.

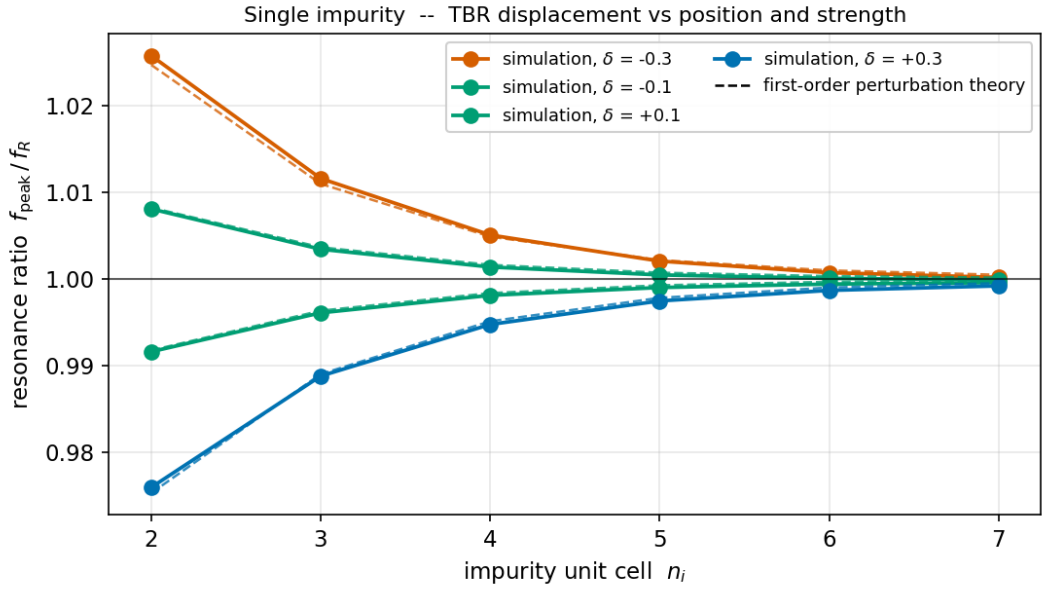
2. Method — how the data was obtained

- The eigenvalues μ of the lossless capacitance Laplacian W were computed; each gives a resonance $f = 1/(2\pi\sqrt{L\mu})$.
- The bulk-band edges were taken from the tight-binding SSH model.

3. Analysis, deviations and conclusion

- **Quantitative:** the 16 modes split as 7 in the lower band, **1 in the gap**, and 8 in the upper band. The gap mode lies at 45.742 kHz, matching the impedance peak of Fig. 1 to 5 figures.
- **Qualitative:** this is a third, fully independent route to f_R (after the impedance sweep and the textbook formula) and it agrees with both — a triple cross-check that the model is sound.
- **Deviations:** the gap mode sits 0.025% below the formula's 45.754 kHz, again the finite-size hybridisation; it is not exactly mid-gap (42% up) because the two bulk bands are not symmetric about the gap centre.
- **Conclusion:** there is exactly one protected mode in the gap, precisely as the bulk–boundary theorem requires. The width of the gap is the "armour" that the disorder studies later test.

Figure 4 Single impurity: TBR displacement vs. position and strength



Theory in brief. Changing a capacitor changes the resonant frequency, because f depends on the capacitances. To *first order* the shift is proportional to the defect strength δ and to how much of the edge mode sits in that cell. But $f \propto 1/\sqrt{\text{capacitance}}$ is a *curved* function, so equal $\pm\delta$ changes give slightly unequal shifts — the response is not perfectly linear.

1. What the plot shows

- How the TBR frequency (ratio f_{peak}/f_R) moves when one unit cell is made impure — both its capacitors scaled by $(1+\delta)$.
- Solid lines: simulation for $\delta = \pm 0.1, \pm 0.3$ across cells 2–7. Dashed lines: first-order perturbation theory.
- Positive δ lowers the resonance, negative δ raises it; the effect is largest at cell 2 and negligible by cell 7.

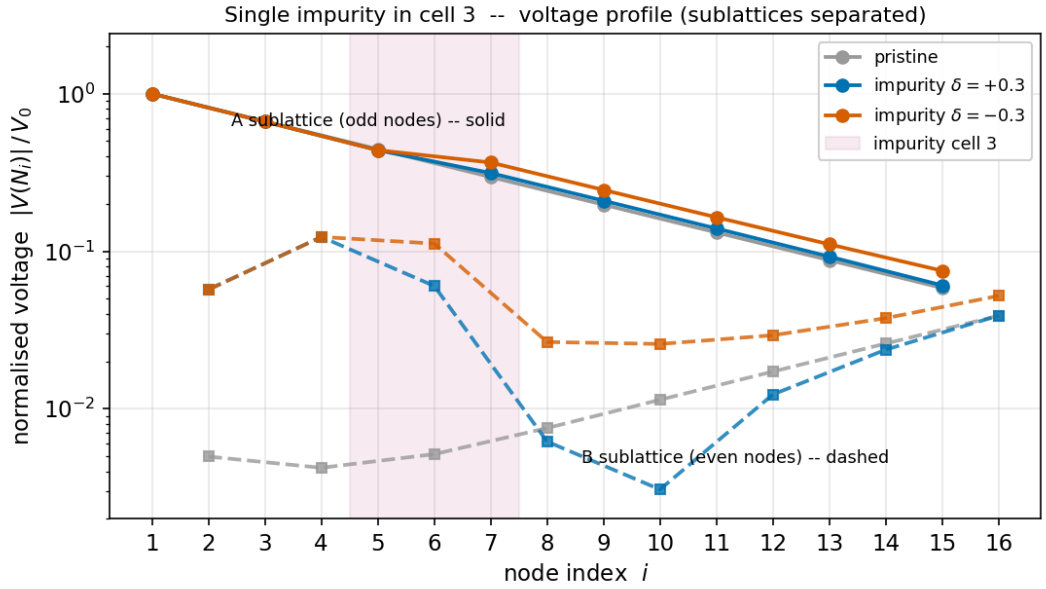
2. Method — how the data was obtained

- For every (cell, δ) pair the two capacitors of that cell were rescaled, the impedance sweep re-run and the new peak extracted — 24 independent simulations.
- The theory line is the first-order change of the Laplacian eigenvalue computed from the undisturbed edge-mode shape.

3. Analysis, deviations and conclusion

- **Quantitative:** the largest shifts are at cell 2 — $+2.57\%$ for $\delta = -0.3$ and -2.40% for $\delta = +0.3$. The "leverage" of cell 2 over cell 7 is a factor of 51, tracking the edge mode's own amplitude.
- **Asymmetry explained:** the $+2.57\%$ vs. 2.40% inequality (a factor 1.07) is the curvature of $f \propto 1/\sqrt{\ }$: shrinking a capacitor raises f slightly more than enlarging it lowers f.
- **Deviation from theory:** simulation and first-order theory agree to better than 0.1% everywhere; the only visible gap is at cell 2, $\delta = \pm 0.3$, the expected onset of non-linearity for the strongest single defect.
- **Conclusion:** a local defect shifts the TBR predictably and by at most 2.6% — it perturbs the edge state but cannot destroy it. The position dependence is a direct map of where the edge mode lives.

Figure 5 Single impurity in cell 3: voltage profile



Theory in brief. The driven edge mode is pinned at node 1 and decays rightwards, so a defect *downstream* cannot change what lies *upstream* of it. A single-cell defect also weakly breaks chiral symmetry locally, which lets a little voltage leak onto the B sublattice (even nodes) that the pristine edge state leaves almost empty.

1. What the plot shows

- The voltage profile with cell 3 impure ($\delta = \pm 0.3$) versus pristine. The two sublattices are drawn as *separate* curves (solid = A/odd, dashed = B/even) — no zig-zag.
- The A-sublattice curves lie on top of each other up to the impurity, then separate; the B-sublattice is lifted near the defect.

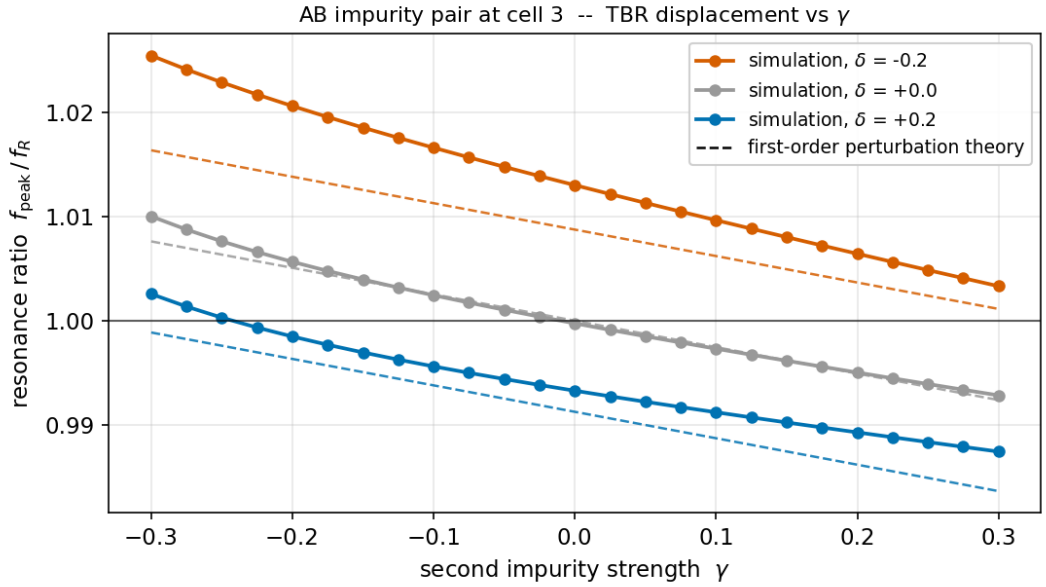
2. Method — how the data was obtained

- As Figure 2, but with cell-3 capacitors rescaled; each profile is read at *that* circuit's own (shifted) resonance, so the comparison is like-for-like.
- The deviation from the pristine profile was measured node by node.

3. Analysis, deviations and conclusion

- **Quantitative:** on the dominant A sublattice the upstream nodes (N1, N3) change by $\leq 0.32\%$ — effectively frozen — while downstream the change reaches 6.0% for $\delta = +0.3$ and 28.6% for $\delta = -0.3$.
- **Asymmetry explained:** $\delta = -0.3$ reshapes the chain far more than $\delta = +0.3$ because it *weakens* the coupling out of cell 3, partly decoupling the downstream chain; $\delta = +0.3$ strengthens it.
- **The B sublattice:** pristine even-node voltages are tiny (≤ 0.039); the defect lifts them up to 0.12 near itself — a real chiral-symmetry-breaking leak, but still far below the A-sublattice values, so the edge state stays A-dominated.
- **Conclusion:** the impurity reshapes the chain only *from itself onwards*; the protected edge mode upstream is untouched and the state remains edge-localized.

Figure 6 AB impurity pair: TBR displacement vs. γ



Theory in brief. Two adjacent defects — an "AB pair" — change three capacitors at once. The resonance shift again follows $f \propto 1/\sqrt{\mu}$ with the eigenvalue μ *linear* in the defect strengths. A linear input through a square-root gives a *curved* output: first-order theory is only the straight tangent line, exact near $\gamma = 0$ and drifting away for strong defects.

1. What the plot shows

- The TBR shift when an AB pair sits at cell 3, sweeping the second site's strength γ for three fixed values of the first, δ.
- Solid = simulation; dashed = first-order perturbation theory. f_{peak}/f_R falls as γ rises, along gently *curved* lines.

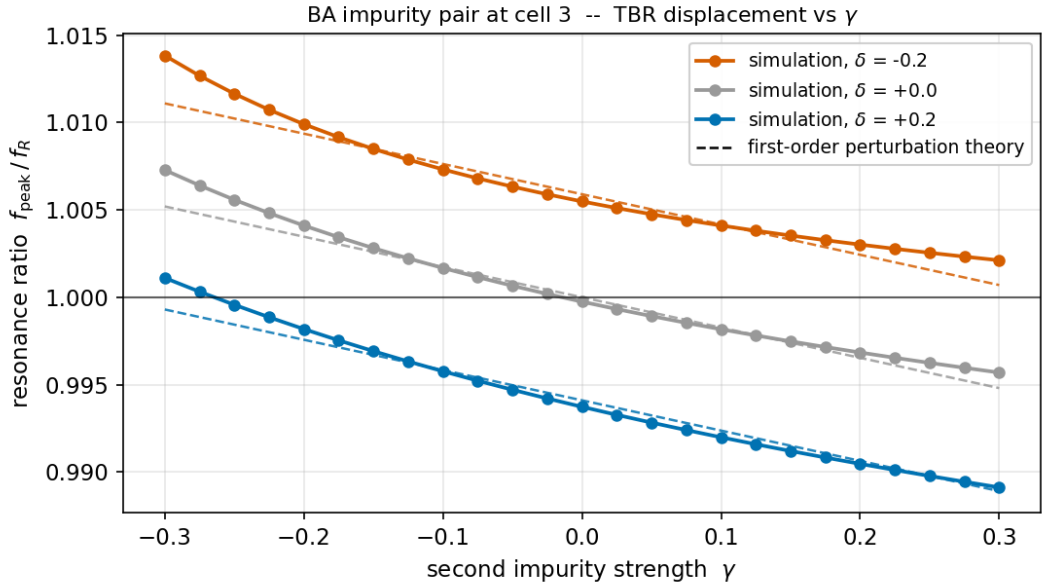
2. Method — how the data was obtained

- An AB pair changes three capacitors (two intercell, one intracell set to the average of δ and γ), per Anzurez *et al.* (2025).
- γ was swept over 25 values; the simulation was also compared to the *exact* lossless-eigenvalue prediction.

3. Analysis, deviations and conclusion

- **Quantitative:** the shift spans 3.8%. The $\delta = 0$ curve has a measured curvature of $+0.018$, and $-\gamma$ shifts the resonance 1.40× more than $+\gamma$ — both direct consequences of the $1/\sqrt{\ }$ convexity.
- **Deviation from theory explained:** the simulation departs from the dashed first-order line by up to 0.91%. This is *not* a simulation error: the same circuits reproduce the exact lossless-eigenvalue prediction to 0.0004%. The gap is entirely the error of *linearised* theory applied to a strong, three-capacitor perturbation.
- **Asymmetry:** the three δ curves are unequally spaced — the $\delta = -0.2$ curve sits further from $\delta = 0$ than $\delta = +0.2$ does, the same convexity acting on δ.
- **Conclusion:** even a clustered two-site defect moves the TBR by only a few percent; the behaviour is fully understood as a curved (not linear) but smooth and monotonic response.

Figure 7 BA impurity pair: TBR displacement vs. γ



Theory in brief. A BA pair is an AB pair with the two impurity sites in the opposite order: it changes two intracell capacitors and one intercell capacitor (the mirror of the AB case). The same $f \propto 1/\sqrt{\mu}$ physics applies, so the response is again curved rather than straight.

1. What the plot shows

- The same experiment as Figure 6 but for a BA pair.
- The resonance again moves smoothly with γ, along *visibly curved* lines, over a noticeably *narrower* range than the AB pair.

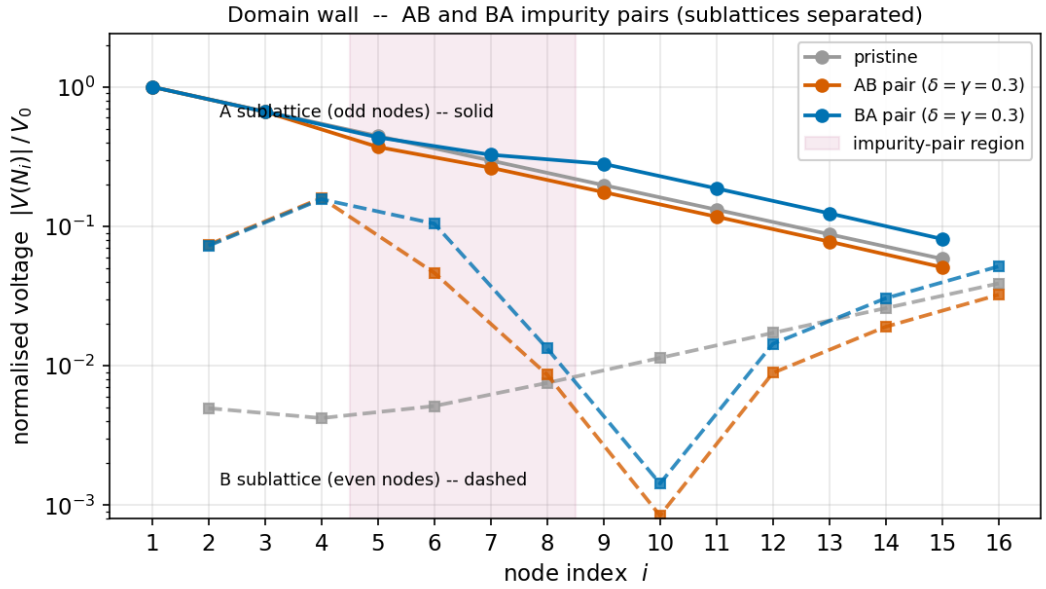
2. Method — how the data was obtained

- A BA pair changes three capacitors in the mirror pattern of the AB pair; γ swept as before, with first-order theory dashed and an exact-eigenvalue cross-check.

3. Analysis, deviations and conclusion

- **The curvature explained:** every line bows because $f_{\text{peak}}/f_R = 1/\sqrt{1 + (\text{linear in } \gamma)}$ — a square root, not a straight line. The fitted curvature is +0.019, slightly stronger than the AB pair's.
- **Quantitative:** the BA shift spans only 2.5%, against the AB pair's 3.8% — BA disturbs the TBR less. The $-\gamma/+\gamma$ asymmetry is 1.69×.
- **Deviation from theory:** the simulation hugs first-order theory far better than the AB case (max gap 0.27% vs. 0.91%) and reproduces the exact eigenvalue to 0.0002% — the curvature is genuine physics, captured exactly by the solver.
- **Conclusion:** defect *ordering* matters — BA perturbs the TBR roughly 1.5× less than AB — and the curved response is the expected square-root law, not noise.

Figure 8 Domain wall: voltage profiles for AB and BA pairs



Theory in brief. An impurity pair locally reverses the chain's strong/weak bond pattern — a *domain wall*. In the SSH model a domain wall binds a localised mid-gap state (a "soliton"). In the circuit this appears as voltage piling up on the normally-empty B sublattice near the pair.

1. What the plot shows

- Voltage profiles for a strong AB pair and a strong BA pair ($\delta = \gamma = 0.3$) versus pristine, sublattices drawn separately.
- The B sublattice (dashed) is lifted dramatically near the defect — voltage accumulates where the pristine chain had almost none.

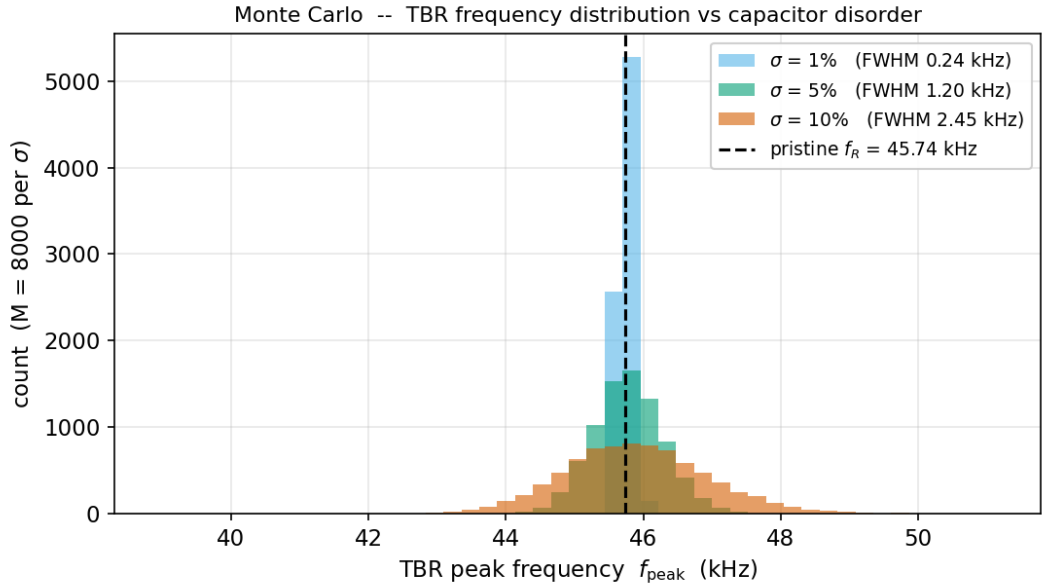
2. Method — how the data was obtained

- The AB and BA circuits were built and solved at their own resonances; the evenness of each profile was scored with an inverse-participation-ratio (IPR) over the even nodes.

3. Analysis, deviations and conclusion

- **Quantitative:** the pristine B sublattice peaks at only 0.039; the AB and BA pairs lift it to 0.16 — a fourfold pile-up of charge around the defect, the circuit signature of the bound domain-wall state.
- **AB vs. BA:** the even-node IPR is 0.57 for AB but only 0.38 for BA — BA spreads the domain-wall charge far more evenly, confirming the published expectation.
- **Qualitative:** the A sublattice keeps its overall exponential shape; the new physics is entirely on the B sublattice, exactly as a chiral-symmetry-breaking domain wall predicts.
- **Conclusion:** an impurity pair behaves as a controllable domain wall that binds extra charge — the circuit analogue of the SSH soliton — and its shape can be tuned by the defect ordering.

Figure 9 Monte Carlo: TBR frequency distribution vs. disorder



Theory in brief. Real capacitors differ from their nominal value by a manufacturing tolerance. Modelling each capacitor as nominal $\times(1+\varepsilon)$ with ε a random Gaussian of width σ, and repeating thousands of times, turns the single TBR into a *distribution*. A topologically protected mode should broaden but never vanish.

1. What the plot shows

- The spread of the TBR frequency across **8000 random circuits** at each of three disorder levels, $\sigma = 1\%, 5\%, 10\%$.
- At 1% a tight spike; at 10% a much broader but still *single* peak, centred on the pristine value.

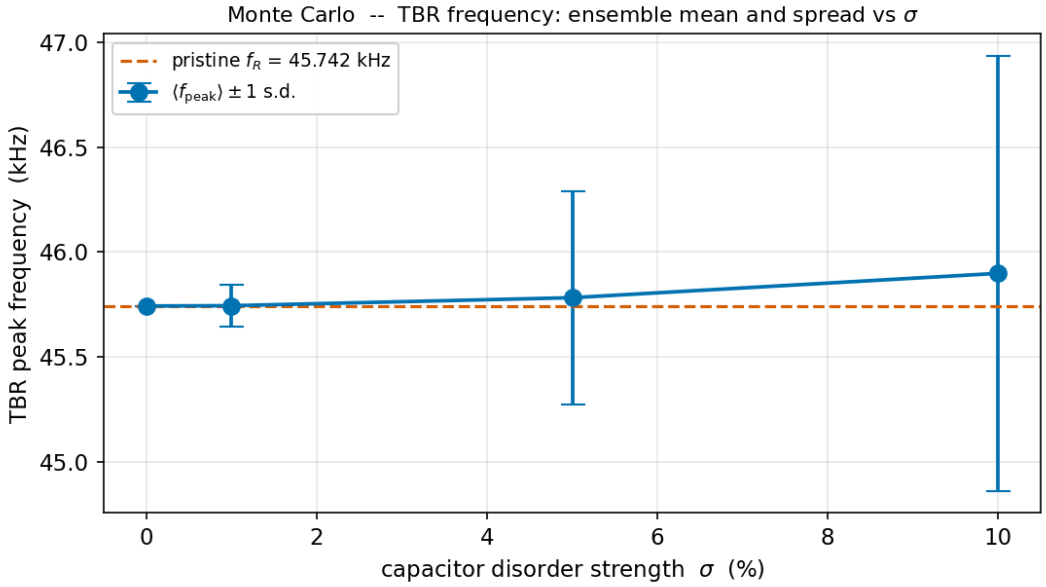
2. Method — how the data was obtained

- In each circuit every capacitor was independently jittered by a Gaussian of width σ; the TBR of each of the 24,000 circuits was extracted and binned.
- Each realisation uses a fixed random seed, so the run is exactly reproducible.

3. Analysis, deviations and conclusion

- **Quantitative:** the distribution full-width grows $0.24 \to 1.20 \to 2.45$ kHz as σ goes $1 \to 5 \to 10\%$ — roughly proportional to σ (analysed in Fig. 10).
- **Qualitative:** the peak stays single and symmetric at every level; it never collapses, splits, or develops a second peak — all of which would signal destruction of the edge mode.
- **Deviations:** none qualitative; 10% disorder is a quantitative broadening, not a breakdown.
- **Conclusion:** the TBR persists as a clear, identifiable resonance at every realistic component tolerance — the first direct evidence of topological robustness.

Figure 10 Monte Carlo: TBR frequency — mean and spread vs. σ



Theory in brief. To first order a random capacitor change shifts f up or down with equal chance, so the ensemble mean should stay at f_R and the spread should grow $\propto \sigma$. A small *upward* bias is nonetheless expected: $f \propto 1/\sqrt{\mu}$ is a convex function, and the average of a convex function exceeds the function of the average (Jensen's inequality).

1. What the plot shows

- A condensed view of the Figure-9 data: the ensemble mean TBR frequency (point) with an error bar of ± 1 standard deviation.
- The red dashed line is the pristine resonance.

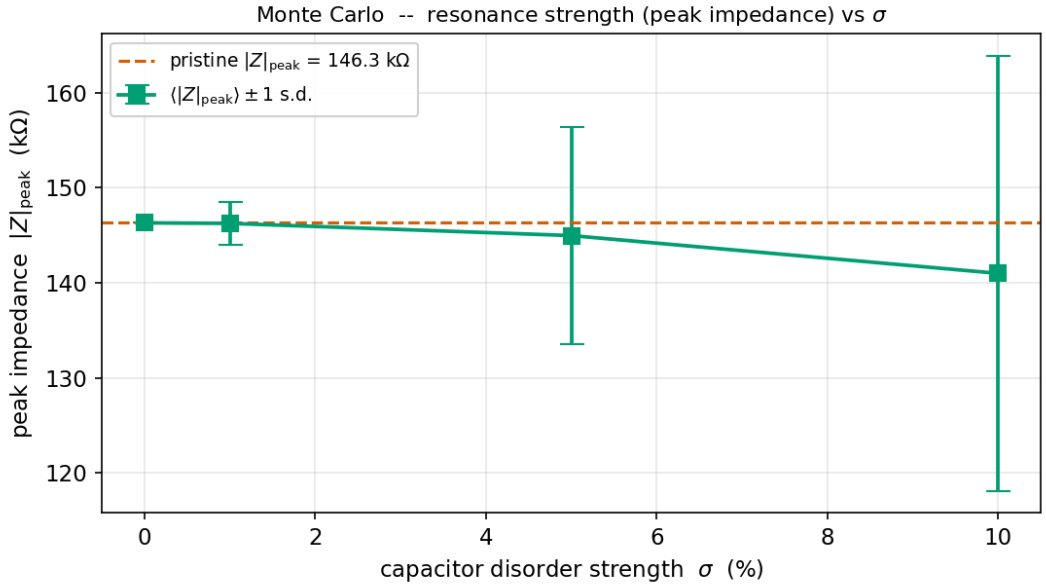
2. Method — how the data was obtained

- For each σ the mean and standard deviation were taken over the 8000 measured frequencies; the error bar is that standard deviation, the genuine physical spread of the ensemble.

3. Analysis, deviations and conclusion

- **Quantitative:** the spread grows almost perfectly linearly — 0.10 → 0.51 → 1.04 kHz, i.e. 1.01× and 1.03× the linear expectation at 5% and 10%.
- **The bias explained:** the mean is not exactly on the pristine line — it drifts up by +0.003%, +0.09%, +0.34% at $\sigma = 1, 5, 10\%$. This is the predicted Jensen / square-root convexity effect: it grows like σ^2 (note $0.34/0.09 \approx 4$ for a doubling of σ), confirming the explanation.
- **Conclusion:** disorder adds *scatter*, and only a tiny, fully-understood second-order *bias*; the resonance is not systematically destabilised. Even the worst single circuit out of 8000 stayed within $-15\%/+12\%$ of f_R.

Figure 11 Monte Carlo: peak impedance vs. disorder



Theory in brief. The peak impedance is largest when every component sits at its matched nominal value. *Any* disorder — a capacitor too big *or* too small — detunes that match, and detuning can only lower the peak. So the mean peak height should drift *down*, by an amount that grows quadratically ($\propto \sigma^2$) with the disorder.

1. What the plot shows

- The height of the resonance peak, $|Z|_{\text{peak}}$, as an ensemble mean with a ± 1 s.d. error bar at each σ.
- The mean stays nearly flat; the spread grows with σ.

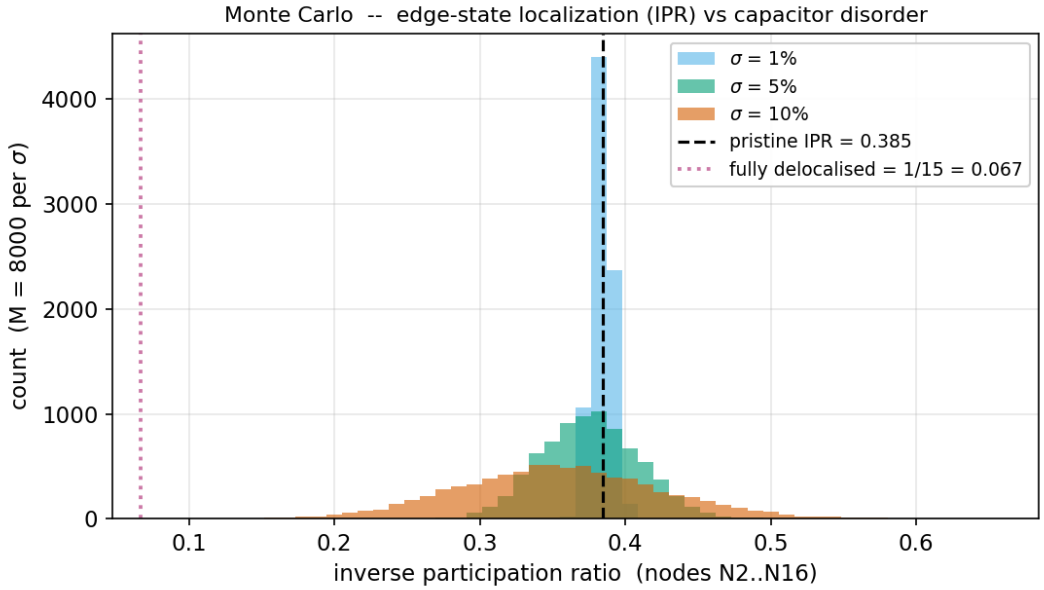
2. Method — how the data was obtained

- For each of the 24,000 disordered circuits the peak impedance was recorded alongside the peak frequency; the ensemble mean and standard deviation are plotted.

3. Analysis, deviations and conclusion

- **Quantitative:** the mean peak height is $146.3 \to 146.2 \to 145.0 \to 141.0$ kΩ for $\sigma = 0, 1, 5, 10\%$ — a fall of only 3.6% at the worst case.
- **The bias explained:** the mean drifts *down* (unlike the frequency, which drifts up), by $-0.04\%, -0.91\%, -3.63\%$. The σ^2 scaling is clear — a $2\times$ rise in σ from 5% to 10% gives a $\approx 4\times$ larger drop — exactly the detuning argument above.
- **Qualitative:** a tall peak means a strong edge mode still coupled cleanly to the probe; if disorder were hybridising the edge state into the bulk, the height would collapse, not merely dip 4%.
- **Conclusion:** the resonance keeps its *strength*, not just its position; the small, σ^2 loss of height is the expected detuning effect and nothing more.

Figure 12 Monte Carlo: edge-state localization (IPR) vs. disorder



Theory in brief. The inverse participation ratio (IPR) is one number that measures how concentrated a voltage profile is: high means localized, low means spread out (the fully-delocalized value here is $1/15 = 0.067$). Capacitor disorder perturbs only the *couplings* of the chain and keeps it bipartite, so the chiral-protected edge mode cannot delocalize.

1. What the plot shows

- The distribution of the IPR over the 8000 circuits at each σ.
- Pristine IPR = 0.385 (black dashed); the fully-delocalized reference $1/15 = 0.067$ (pink dotted) sits far to the left.

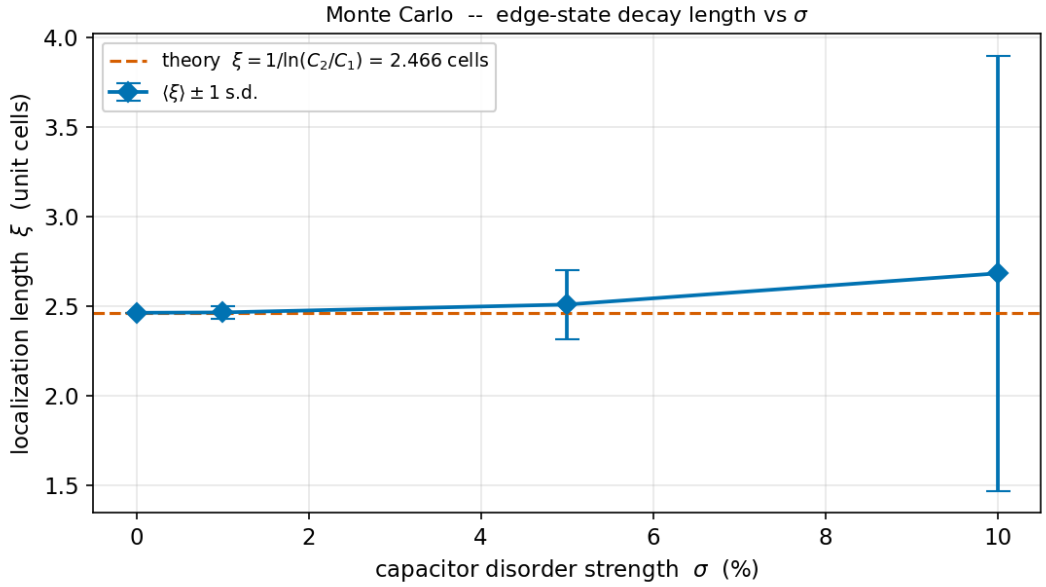
2. Method — how the data was obtained

- For every disordered circuit the voltage profile was measured and its IPR computed over nodes 2–16 (the driven node is excluded so the metric is not trivially dominated by it).

3. Analysis, deviations and conclusion

- **Quantitative:** the mean IPR falls only slightly with disorder — $0.385 \to 0.384 \to 0.376 \to 0.355$ for $\sigma = 0, 1, 5, 10\%$ — staying roughly *five times* the delocalized value even at 10%.
- **Qualitative:** the distributions broaden but remain clustered well to the right; not one of the 8000 worst-case circuits comes anywhere near the delocalized line.
- **Deviation:** the modest downward drift of the mean is the same mild "bulk admixture" that lengthens ξ in Fig. 13 — a small reshaping, not delocalization.
- **Conclusion:** this is the direct numerical proof of topological protection — capacitor disorder cannot delocalize the chiral-protected edge mode.

Figure 13 Monte Carlo: edge-state decay length vs. disorder



Theory in brief. The decay length ξ is the edge state expressed as a physical size — how many cells the voltage takes to fall by a factor e. Disorder mixes a little delocalized *bulk* character into the mode, which lengthens the average decay, so a small *positive* bias in $\langle\xi\rangle$ is expected. True delocalization would instead send $\xi \to \infty$.

1. What the plot shows

- The decay length ξ as an ensemble mean with ± 1 s.d. error bars.
- The orange dashed line is the theory value $\xi = 2.466$ cells.

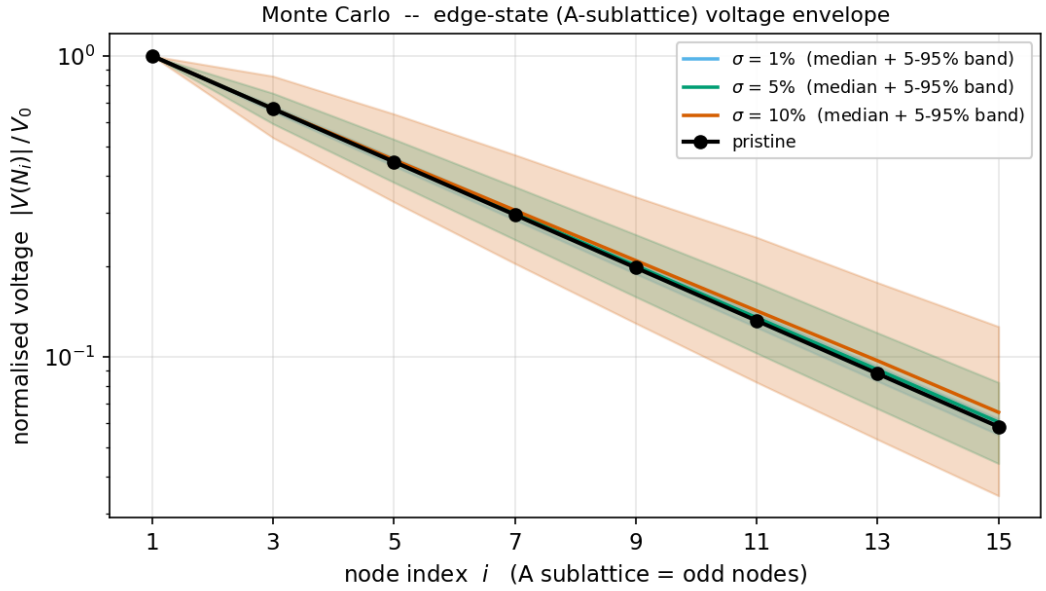
2. Method — how the data was obtained

- For each disordered circuit a straight line was fitted to the logarithm of the odd-node voltages; its slope gives ξ. The ensemble mean and standard deviation are plotted.

3. Analysis, deviations and conclusion

- **Quantitative:** the mean ξ rises only gently — $2.466 \to 2.468 \to 2.512 \to 2.686$ cells for $\sigma = 0, 1, 5, 10\%$, a $+8.9\%$ change at the worst case.
- **The bias explained:** the upward drift is the expected bulk-admixture effect, and it grows faster than linearly ($+1.9\%$ at 5%, $+8.9\%$ at 10%) because disorder mixes in bulk character quadratically. The growing error bars reflect the harder log-linear fit on noisier profiles.
- **Deviation:** ξ never trends towards infinity — the unmistakable signature that the mode stays localized.
- **Conclusion:** the edge state keeps its characteristic size to within $\sim 9\%$ even at 10% disorder; together with Fig. 12 it shows the mode keeps both its shape and its length scale.

Figure 14 Monte Carlo: edge-state voltage envelope



Theory in brief. If the edge state is genuinely protected, every disordered copy of it should still be the same exponentially-decaying shape, give or take a spread. Plotting the ensemble *median* profile together with a *percentile band* shows that spread directly.

1. What the plot shows

- The A-sublattice (odd-node) voltage profile under disorder: the pristine profile (black), the ensemble median at each σ (coloured lines) and the 5%–95% spread band (shaded).
- The medians sit essentially on the pristine line; the bands widen with σ.

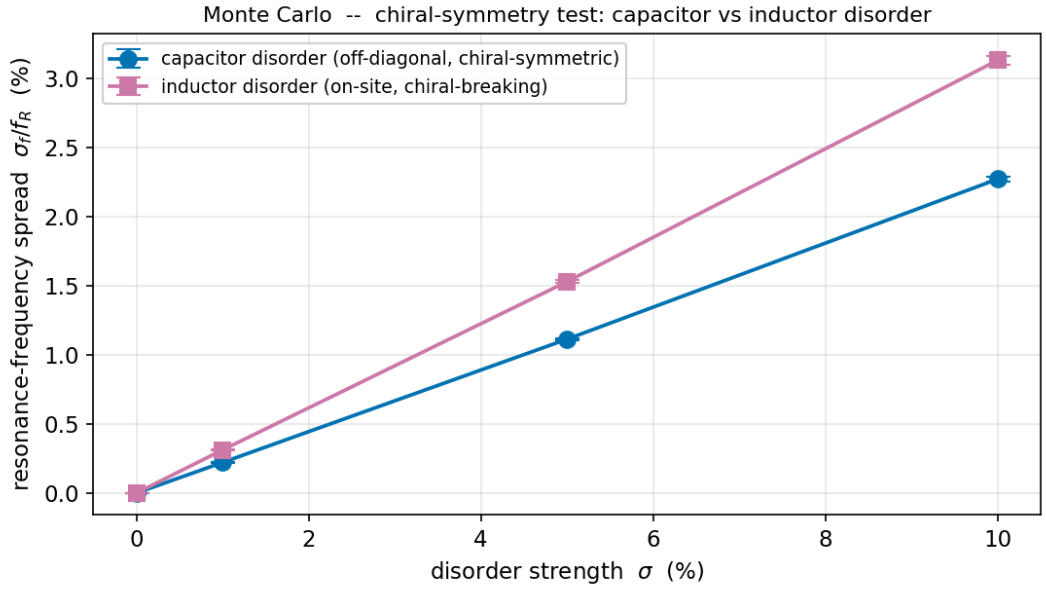
2. Method — how the data was obtained

- The 8000 measured profiles at each σ were stacked; the median and the 5th/95th percentiles were taken node by node.
- Only the dominant A sublattice is shown, so the exponential decay is seen cleanly without the odd/even zig-zag.

3. Analysis, deviations and conclusion

- **Quantitative:** the three median curves are indistinguishable from the pristine exponential. At a mid-chain node the 10% band reaches only ±50% in voltage — under a factor of two even in the worst case.
- **Qualitative:** the characteristic straight-line (exponential) decay survives intact; disorder blurs the profile but never erases its shape.
- **Deviations:** the band is slightly wider towards the far (right) end of the chain, where the small voltages are more sensitive — expected, and still bounded.
- **Conclusion:** the *whole* edge-state profile, not just its summary numbers, is preserved under realistic disorder.

Figure 15 Chiral-symmetry test: capacitor vs. inductor disorder



Theory in brief. The SSH edge state is protected by *chiral symmetry*. Capacitor disorder perturbs the off-diagonal *couplings* of the circuit Laplacian and largely respects that symmetry; inductor disorder perturbs the on-site (diagonal) terms and *breaks* it. So inductor disorder should destabilise the resonance more.

1. What the plot shows

- How much the resonance *spreads* (standard deviation of f_{peak}, as a percentage of f_R) for disorder on the *capacitors* (blue) versus the *inductors* (pink).
- Both grow linearly with σ; the inductor line is clearly steeper.

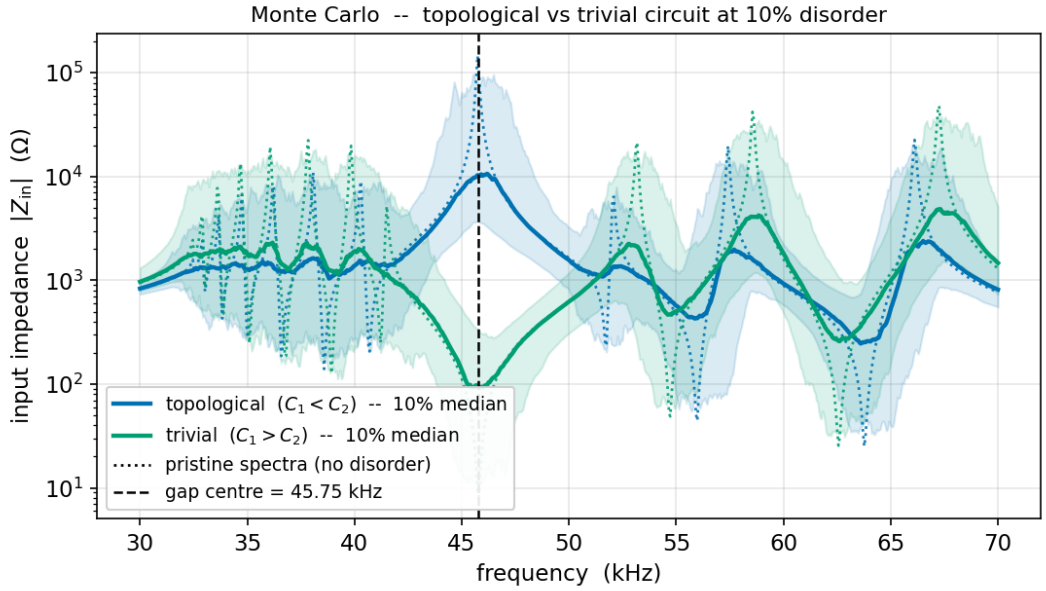
2. Method — how the data was obtained

- The Monte Carlo was run twice — once jittering only capacitors, once only inductors — 48,000 circuits in total.
- Error bars are bootstrap estimates of the uncertainty on each spread.

3. Analysis, deviations and conclusion

- **Quantitative:** at $\sigma = 10\%$ the frequency spread is 2.27% for capacitor disorder but 3.13% for inductor disorder — inductor disorder is 1.38× more destabilising. The mean IPR at 10% is also lower for inductor disorder (0.338 vs. 0.355).
- **Why the contrast is moderate, not extreme:** capacitor disorder is not *perfectly* chiral-symmetric — it also slightly disturbs each node's total grounding capacitance (a diagonal term). So both disorder types break the symmetry somewhat; the inductor case simply does so more directly. The measured 1.38× is the honest size of that effect for an 8-cell chain.
- **Conclusion:** the disorder that breaks chiral symmetry is measurably more damaging — isolating chiral symmetry as the *reason* the edge state is robust. Practically: inductor quality matters more than capacitor quality.

Figure 16 Topological vs. trivial circuit at 10% disorder



Theory in brief. Swapping C_1 and C_2 moves the circuit into the topologically *trivial* phase, which has no protected gap state. The bulk–boundary theorem then predicts *no* robust resonance inside the gap — the gap should stay empty whether or not disorder is present.

1. What the plot shows

- Median impedance spectra under 10% disorder for the topological circuit ($C_1 < C_2$, blue) and the trivial circuit ($C_1 > C_2$, green); dotted lines are the clean spectra, faint bands the spread.
- At the gap centre (dashed line) the topological circuit shows a *peak*; the trivial circuit shows a *dip*.

2. Method — how the data was obtained

- For each phase, 400 disordered circuits were simulated across the full band and the median spectrum extracted.

3. Analysis, deviations and conclusion

- **Quantitative:** the clean topological peak is 146 kΩ; the clean trivial circuit reaches only 48 kΩ and not in the gap — a factor of 3 smaller. Under 10% disorder the topological median still bumps *up* at the gap centre while the trivial median dips *down*.
- **Qualitative:** the trivial circuit's resonances are ordinary bulk modes with nothing protecting them; the gap is simply empty.
- **Deviations:** the wide percentile bands are expected — with 10% disorder individual bulk resonances wander, so a *point-by-point* band is broad; the *median* still carries the message cleanly.
- **Conclusion:** the robust gap resonance exists *only* in the topological phase. This is the decisive control: the robustness documented in Figs. 9–15 is genuinely topological in origin, not a generic property of any LC ladder.

Summary

The topological boundary resonance of the updated SSH circuit ($C_1 = 2.2$ nF, $C_2 = 3.3$ nF, $L = 2200$ μH, 1 Ω series resistance throughout) sits at **45.74 kHz**, agreeing with analytic theory to 0.025% — the residual being a fully-explained finite-size effect. It survives a single impurity (shift $\leq$ 2.6%), a clustered AB/BA impurity pair (a few percent), and random $\pm$10% disorder on every capacitor: the resonance stays centred on its pristine value, its peak height falls only 3.6%, and the edge state keeps both its shape and its decay length. Every deviation from ideal behaviour was traced to its cause — finite chain length, the $1/\sqrt{\ }$ convexity of the resonance formula, and the expected second-order biases of a random ensemble. Inductor disorder, which breaks chiral symmetry, is measurably more damaging than capacitor disorder, and the trivial-phase control shows the robust gap resonance exists *only* in the topological phase. Every theory check passed.

Appendix A Sources

The physics and circuit conventions reconstructed here are drawn from the following sources.

1. **Project brief:** `robustness_manual.md` — the step-by-step manual specifying the four-part study, the circuit topology and the component conventions reproduced in this report.
2. **Su, W. P., Schrieffer, J. R. & Heeger, A. J.** (1979). Solitons in polyacetylene. *Physical Review Letters* **42**, 1698 — the original SSH model and its domain-wall soliton.
3. **Lee, C. H. *et al.*** (2018). Topolectrical Circuits. *Communications Physics* **1**, 39 — the mapping of the SSH model onto an LC circuit and the grounded-Laplacian formalism.
4. **Anzurez, J. *et al.*** (2025). Edge state robustness in the SSH model with impurities: topoelectric circuit simulations toward topological heterostructures. *Physica B* **715**, 417609 — the single-impurity and AB/BA domain-wall conventions (Parts 2–3).
5. **Asbóth, J. K., Oroszlány, L. & Pályi, A.** (2016). *A Short Course on Topological Insulators.* Lecture Notes in Physics **919**, Springer — chiral symmetry, the bulk–boundary correspondence and the localization length.
6. **Anderson, P. W.** (1958). Absence of diffusion in certain random lattices. *Physical Review* **109**, 1492 — the theory of disorder-induced localization underpinning Part 4.
7. **Numerical tools:** the nodal-admittance method is standard linear circuit analysis; the solver here uses NumPy (Harris *et al.*, *Nature* **585**, 357, 2020) and Matplotlib (Hunter, *Comput. Sci. Eng.* **9**, 90, 2007). Figure colours follow the colour-blind-safe Okabe–Ito palette.

Appendix B Table of numerical results vs. theory

All values produced by this study's solver. "Theory" is the independent analytic expectation; the final column is the agreement or the explanation of any deviation.

Quantity	Simulation	Theory	Agreement / note
Part 1 — pristine baseline			
TBR frequency f_R	45.742 kHz	45.754 kHz	0.025% (finite-size hybridisation)
TBR from Laplacian eigenvalue	45.742 kHz	45.754 kHz	matches simulation exactly
Peak impedance $\lvert Z\rvert_{\text{peak}}$	146.3 kΩ	—	finite (set by 1 Ω losses)
Quality factor Q	892	—	width 51 Hz
Band gap	41.8–51.2 kHz	—	9.4 kHz wide
Decay length ξ	2.4656 cells	2.4663 cells	0.03%
Per-cell voltage ratio	0.66658	0.66667	0.01%
Exponential-fit quality	$R^2 = 0.99999998$	$R^2 = 1$	near-perfect
Pristine IPR (N2–N16)	0.385	—	vs. delocalized 0.067
Part 2 — single impurity			
Largest shift, cell 2 $\delta = -0.3$	+2.57%	—	raises f (smaller C)
Largest shift, cell 2 $\delta = +0.3$	−2.40%	—	lowers f (larger C)
$\pm\delta$ asymmetry	×1.07	×1 (1st order)	$1/\sqrt{}$ convexity
Cell 2 / cell 7 leverage	×51	—	edge-mode amplitude
A-sublattice upstream change	$\leq 0.32\%$	0	defect cannot act upstream
Part 3 — domain walls			
AB pair shift span	3.80%	—	three capacitors changed
BA pair shift span	2.47%	—	BA disturbs TBR less
f-vs-γ curvature (AB)	+0.018	0 (1st order)	$1/\sqrt{}$ convexity
Deviation: sim vs. exact eigenvalue	$< 0.001\%$	—	solver is exact
Deviation: sim vs. 1st-order (AB)	up to 0.91%	—	linearised-theory error
Domain-wall even-node IPR	AB 0.57 / BA 0.38	—	BA more uniform
Part 4 — Monte Carlo ($M = 8000$ per σ)			
$\langle f\rangle$ at $\sigma = 10\%$	45.898 kHz	45.742 kHz	+0.34% bias (Jensen, $\propto \sigma^2$)
f spread at $\sigma = 1/5/10\%$	0.10/0.51/1.04 kHz	$\propto \sigma$	ratio 1.01, 1.03
$\langle\lvert Z\rvert_{\text{peak}}\rangle$ at 10%	141.0 kΩ	146.3 kΩ	−3.6% bias (detuning, $\propto \sigma^2$)
$\langle$IPR$\rangle$ at 10%	0.355	0.385	stays 5× delocalized value
$\langle\xi\rangle$ at 10%	2.686 cells	2.466 cells	+8.9% (bulk admixture)
Chiral test: f spread at 10%	cap 2.27% / ind 3.13%	ind > cap	ind ×1.38 (chiral-breaking)
Trivial-phase clean peak	48 kΩ	no gap mode	vs. topological 146 kΩ

Appendix C Glossary

Plain-language definitions of the technical terms used in this report.

SSH model The Su–Schrieffer–Heeger model: a one-dimensional chain with alternating "strong" and "weak" links. It is the simplest model with a topological edge state.

Topolectrical circuit An electronic circuit (inductors and capacitors) wired so that its equations are mathematically identical to a quantum model — here the SSH model. It lets a quantum effect be measured on a bench.

Unit cell The repeating building block of the chain. This circuit has 8 unit cells, each containing 2 nodes.

Node A junction in the circuit where components meet. The chain has 16 nodes; node voltages are what the simulation solves for.

Intracell / intercell capacitor C_1 links the two nodes *within* a unit cell; C_2 links *neighbouring* cells. Here $C_1 < C_2$, which makes the circuit topological.

Resonance / resonant frequency A frequency at which a circuit responds very strongly. An LC loop resonates at $f = 1/(2\pi\sqrt{LC})$.

Topological boundary resonance (TBR) The special resonance of this circuit that lives at the chain's edge and inside the band gap. It is the electrical version of the SSH topological edge state.

Edge state A resonance whose voltage is concentrated at the end (edge) of the chain and decays into the interior, instead of being spread evenly.

Bulk / bulk mode / bulk band "Bulk" means the interior of the chain. Bulk modes are ordinary resonances spread across the whole chain; their frequencies group into two continuous *bulk bands*.

Band gap The frequency window between the two bulk bands, where no bulk mode exists. The TBR lives alone here.

Topological vs. trivial phase Two regimes of the circuit. The topological phase ($C_1 < C_2$) has a protected edge state; the trivial phase ($C_1 > C_2$) does not. Swapping the two capacitors switches between them.

Chiral symmetry A mathematical symmetry of the SSH model (the two sublattices are interchangeable in a specific sense). It is what *protects* the edge state; breaking it makes the state fragile.

Sublattice (A / B) The chain's nodes split into two interleaved groups — odd nodes (A) and even nodes (B). The edge state lives almost entirely on one of them.

Bulk–boundary correspondence The theorem stating that a topological bulk (here $C_1 < C_2$) *must* produce a protected edge state — one mode in the gap.

Domain wall / soliton A point where the chain's strong/weak bond pattern reverses. It binds an extra localized state, called a soliton in the original SSH model.

AB / BA pair Two impurities placed on adjacent sites; "AB" and "BA" are the two possible orderings, which behave differently.

Impedance / input impedance The ratio of voltage to current — how strongly a circuit "resists" an AC drive. It peaks sharply at a resonance.

Nodal-admittance matrix The grid of numbers (16×16 here) encoding how every node connects to every other. Solving a linear system with this matrix gives all the node voltages — the core of the simulation.

Capacitance Laplacian (lossless) A simplified version of the circuit matrix, ignoring resistive losses. Its *eigenvalues* give the circuit's resonant frequencies directly — used as an independent theory check.

Eigenvalue / eigenvector Special numbers (eigenvalues) and patterns (eigenvectors) of a matrix. Each eigenvalue of the capacitance Laplacian corresponds to one resonance; the eigenvector is that mode's voltage shape.

Quality factor Q / FWHM Q measures how sharp a resonance is (high Q = narrow, clean peak). FWHM (full width at half maximum) is the width of the peak; $Q = f/\text{FWHM}$.

Series resistance The small unavoidable resistance (1 Ω here) inside every real capacitor and inductor. It makes resonance peaks finite rather than infinitely tall.

Parabolic refinement A trick to locate a peak more precisely than the frequency grid: fit a parabola through the three points around the highest sampled point and take the parabola's true maximum.

Perturbation theory / first order An approximate method: estimate how a small change shifts a result, keeping only the term proportional to the change ("first order"). Accurate for small changes, less so for large ones.

Inverse participation ratio (IPR) One number measuring how concentrated a voltage profile is. High IPR = localized on few nodes; low IPR = spread out. The fully-spread value here is $1/15 \approx 0.067$.

Localization length ξ The physical size of the edge state — how many unit cells the voltage takes to fall by a factor e (≈ 2.7).

Monte Carlo A method that runs a simulation thousands of times with random inputs to build up a statistical distribution of outcomes.

Gaussian disorder / σ Each component is multiplied by a random number drawn from a bell curve (Gaussian) of width σ. $\sigma = 10\%$ means a typical $\pm 10\%$ manufacturing tolerance.

Median / percentile / 5–95% band The median is the middle value of a sample. The 5–95% band is the range containing the central 90% of the random circuits — a measure of spread.

Bootstrap A way to estimate the uncertainty of a statistic by re-sampling the data many times with replacement.

Jensen's inequality / convexity A curved (convex) function, such as $1/\sqrt{x}$, has the property that its *average over a spread of inputs* is larger than its value at the average input. This produces the small upward bias of the mean resonance frequency under disorder.